

Language Models Can Read Each Other’s Memories

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Abstract

When one language model hands work to another, it writes a message, and everything it computed while reading is left behind. We ask whether a model can hand over its memory instead: the key–value cache it built while reading. Between two frozen models of different size, we learn a map that translates the small model’s cache into the large model’s, and the large model reads the result as if it were its own. Transplanted memories turn out to be *readable*: the receiver answers questions about text it never saw, at 63% on a binding test where chance is 12.5%, and at 53.0 F1 on SQuAD against 54.7 when it reads the passage itself. They are *believed*: handed another item’s memory, the receiver answers from it 64% of the time. They *carry doubt*: when the sender’s evidence weakens, the receiver’s confidence follows through the memory (Spearman .28) and not through another item’s memory (.00), while the words “confidence: low” barely move it. And they are *portable, within limits*: the map crosses model families about as well as it works inside one, and fails for a sibling that differs in vocabulary, depth and head size. We release the maps, the tests, and a two-line audit for any handoff between models: score the message, then score the reader.

1. Introduction

When one person hands a problem to another, they explain what they know, and the explanation is always smaller than the knowing. The alternatives they weighed, how sure they were, the detail they noticed and did not mention: all of it stays behind. Language models working together do the same thing. A small model reads a document and writes an answer or a summary; a larger model reads that text and carries on. Routers, cascades and multi-agent systems all

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Preliminary work. Under review by the International Conference on Machine Learning (ICML). Do not distribute.

hand work over in words.

A model is not a person, though. What it remembers about a document is not a feeling. It is a tensor, the key–value (KV) cache its attention layers build while reading, and a tensor can be copied. This paper asks a plain question: *can one language model read another’s memory?*

We take two frozen models of different size, K2-Horizon-3.7B and K2-Horizon-7B, and learn a map from the small model’s cache to the large model’s. The small model reads a passage. The large model never sees the passage; it receives the translated cache in place of its own and answers questions about what is in it. We then ask four things about these transplanted memories, and the answers organise the paper.

Readable (Section 3). The receiver answers from a memory it never formed. Told eight made-up names and their colours only through the cache, it names the right colour 63% of the time, where chance is 12.5%. On SQuAD it reaches 53.0 F1, against 54.7 when it reads the passage, and on two-paragraph questions the transplanted memory supplies the paragraph the receiver is missing.

Believed (Section 4). Hand the receiver another item’s memory and it answers from that memory 64% of the time. It is reading, not guessing, and a map trained with nothing to carry cannot do the same. The memory is read in the receiver’s later layers: the first half alone carries nothing usable.

Carrying doubt (Section 5). A model’s answer is often worth less than how sure it is. When we weaken the sender’s evidence, removing the sentence that answers the question, the receiver’s confidence moves with the sender’s through the memory, and does not move through another item’s memory. Through words it barely moves: appending “confidence: low” to the sender’s answer leaves the receiver tracking the sender about as poorly as with no message at all, although the label itself predicts the sender almost perfectly. The signal is in the message; this reader drops it. The effect through the memory is modest, and we report it with the controls and intervals it needs.

Portable, within limits (Section 6). The same recipe carries a memory from Qwen3-4B into K2-Horizon-7B, across model families, about as well as it works inside one; it fails

for a smaller K2 sibling. And memory is heavy: a page of text is 72 MiB of cache, cheap to hand over inside one machine and expensive across a network.

That memory might move between minds has a long and odd history in neuroscience, down to the transfer of a learned sensitisation between sea slugs by injected RNA (Bédécarrats et al., 2018). Between language models it is a matter of engineering. What it implies is less ordinary. If the systems we are now building are, in Minsky’s phrase, societies of mind (Minsky, 1986), the protocol between their members need not be ours. Shannon defined communication as reproducing at one point a message selected at another (Shannon, 1948). We ask what is reproduced when the message is a state rather than a string, and find that more crosses than the words would carry.

2. Transplanting a Memory

Models. The sender is K2-Horizon-3.7B and the receiver K2-Horizon-7B, both frozen and in bf16. They share a tokenizer and the shape of their cache: 36 layers, 8 KV heads, head dimension 128. We record the sender’s keys (before rotary position encoding) and values at every layer while it reads, map them, apply the receiver’s rotary encoding, and insert the result as the receiver’s cache for those positions. The receiver then reads only the question.

The map. It has two parts. First, for each receiver layer and KV head, a ridge regression from the sender’s concatenated keys and values (256 dimensions per head) to the receiver’s, drawing on the three sender layers whose keys and values correlate best with that receiver layer. Second, a gated residual network on top (bottleneck 128, 151 M parameters), trained for 1,500 steps on FineWeb-Edu to lower the receiver’s loss on the next 512 tokens when it reads the mapped cache of the previous 512. Both language models stay frozen. Nothing in training mentions a question, an answer or a test.

Tests. *Binding:* each episode gives eight made-up names one of eight colours (“Noma is amber. Vela is indigo. ...”); only the sender reads it, and the receiver is asked the colour of one name. *Passages:* SQuAD v1.1, where the sender reads the passage and the receiver answers the question greedily; we report token-F1. *Doubt:* SQuAD passages in three versions (clean, contradicted, answer sentence removed), described in Section 5. Every test has the same set of arms: the receiver with the question only; the receiver reading the text itself; the receiver’s own cache re-inserted (a check on the insertion, which matches reading the text to 10^{-4}); the sender’s cache inserted with no map; the mapped cache; and the mapped cache of a *different* episode, which shares the question and the layout but not the content. That last arm is the one that matters. Where it is given another

Table 1. Binding transfer, 8 names $\times$ 8 values, $n = 1000$ episodes, chance 12.5%. All entries are percentages. “follow” is the rate at which the receiver, handed the deranged cache, answers with the other episode’s value; “content” is project – derange.

projector	none	text	raw	project	derange	follow	content
ridge top-3 (closed form)	10.6	100	14	21.7	11.4	18.2	10.3
+ residual, 1500 steps	10.6	100	14	63.0	4.6	63.7	58.4
residual, deranged sender	10.6	100	14	23.4	11.3	22.7	12.1

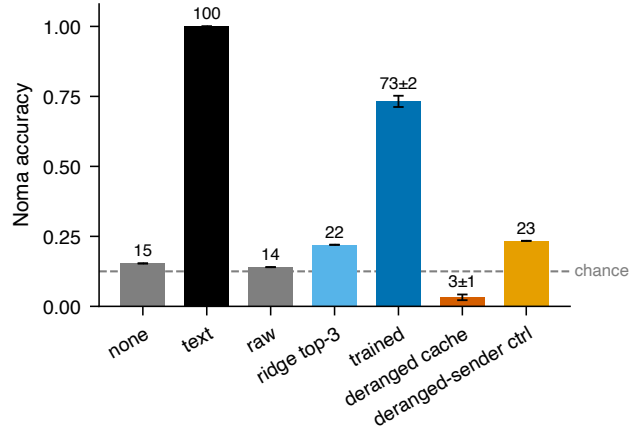


Figure 1. Noma accuracy by arm on the 6-name variant ($n = 300$). “trained” and “deranged cache” are the mean over three training seeds with error bars showing one standard deviation; the deranged-sender control is the *project* arm of the projector trained with a deranged sender ($n = 1000$, 8-name variant). Dashed line: chance (12.5%).

episode’s memory, we also report how often the receiver answers with that episode’s value: the *follow rate*.

3. Transplanted Memories Are Readable

Facts the receiver never saw. On the binding test (Table 1) the receiver names the right colour 63.0% of the time from the transplanted memory alone; chance is 12.5%, and the receiver with only the question gets 10.6%. The sender’s raw cache, inserted with no map, gets 14%: matching the shape of the cache is not enough. The closed-form ridge map on its own gets 21.7%. Most of the transfer comes from the learned part, and from letting the linear map move while it learns: on an easier six-name version of the test, freezing the ridge costs about 14 points and feeding each receiver layer from the matching sender layer, rather than the three best-correlated ones, costs about 40 (Appendix C). On that version three training seeds give 71.3, 73.0 and 75.3% (Figure 1); with cities and animals in place of colours, 61.8 and 67.2%; and moving the fact up to 512 tokens away from the question does not lower accuracy (Appendices D and F).

Real passages. The binding test is synthetic; SQuAD is not. The map was never trained on questions, yet the receiver answering from the sender’s memory of a passage reaches

Table 2. Real passages: SQuAD v1.1 validation, $n = 300$ (600 for the trained projector). Token-F1 against the gold spans; mean gold log-probability per token.

receiver sees	none	text	raw	ridge	trained	deranged-sender ctrl	trained, derange arm
F1	16.6	54.7	0.6	14.3	53.0 (51.8)	15.1	13.8
gold log-prob / token	-3.57	-0.90	-11.2	-3.88	-1.74	-3.77	-4.18

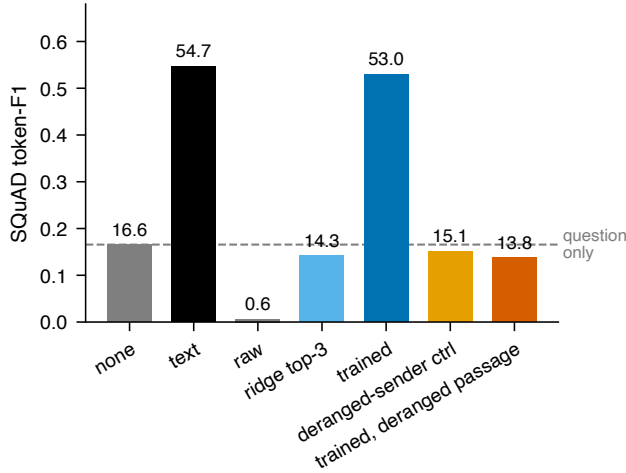


Figure 2. SQuAD token-F1 by arm ($n = 300$). Dashed line: question only. The trained projector sits within two points of the receiver reading the passage itself; the ridge map, the deranged-sender control, and the trained projector given the wrong passage’s cache all sit at or below question-only.

53.0 F1, against 54.7 when it reads the passage itself and 16.6 with the question alone (Table 2 and Figure 2). It recovers almost all of what reading the passage would have given it.

The paragraph you are missing. The cleanest use of another model’s memory is when it holds something you do not. On two-paragraph HotpotQA questions we give each model one paragraph. The receiver with its own paragraph scores 30.1 F1; with both paragraphs as text, 45.8; with its own paragraph plus the sender’s memory of the other, 39.9 ± 2.1 over three seeds, recovering about three fifths of the gap. Another item’s memory leaves it at 29.9–32.1 (Appendix E).

4. Transplanted Memories Are Believed

A receiver that scores well could still be doing something duller than reading: a learned map can act as a prompt that makes the receiver better at answering in general, and a recent audit of relayed caches found that most of a large gain was of that kind (Cheng et al., 2026). Three controls rule this out here.

It follows the wrong memory. Hand the receiver the mapped memory of another episode, with the same names

Table 3. Continuation retention is the wrong headline. Retention $= (\text{loss}_{\text{none}} - \text{loss}_{\text{project}}) / (\text{loss}_{\text{none}} - \text{loss}_{\text{oracle}})$ on 128 fresh held-out sequences; the second column swaps the sender’s text at test time.

projector	retention	retention, sender text deranged
ridge top-3	-0.30	-0.66
+ residual	1.23	0.31
residual, deranged sender	0.77	0.72

Table 4. Where the binding lives. This episode’s mapped cache is injected only into a band of receiver layers, the deranged episode’s elsewhere. 8-name variant, $n = 500$ per band; “all” is the full mapped cache on the same test, $n = 1000$; accuracy in percent, chance 12.5.

layers	0–5	0–11	0–17	12–23	24–35	30–35	18–35	all
accuracy	7.2	8.0	10.4	28.6	30.8	7.6	51.8	63.0

in the same order and a different colour for the one it is asked about, and its accuracy falls to 4.6%, below chance. In 63.7% of episodes it answers with the *other* episode’s colour (Table 1). It is not ignoring the memory and falling back on a guess. It reads the memory and believes it.

A map with nothing to carry fails. We train the identical map with the sender reading text unrelated to what the receiver must predict, so the only thing it can learn is a prompt. It reaches 23.4% on the binding test, about where the closed-form map alone sits. On SQuAD it scores 15.1 F1, at the level of the question alone. Yet by the usual headline for such maps, how much of the loss gap to reading the text it recovers, it scores 0.77, and 0.72 of that survives when its input is swapped for unrelated text (Table 3). A number that rates a content-free prompt at three quarters of reading the text cannot tell a channel from a prompt; the binding test with its wrong-memory arm can.

It is read late. If we give the receiver this episode’s memory in only some layers and another episode’s in the rest, the first half of the network carries nothing usable on its own: layers 0–17 give 10.4%, near chance. The second half alone gives 51.8%, and no band of six layers suffices (Table 4 and Figure 3b). The receiver reads a transplanted fact out of a distributed set of its later layers.

5. Transplanted Memories Carry Doubt

A model’s answer is often worth less than how sure it is. A cascade that escalates when the small model is unsure, a planner that checks before it acts, a buyer deciding whether to trust a seller: all of them need the sender’s doubt, not only its best guess. Text hands over the guess. We ask whether the memory hands over the doubt.

Table 5. Sender uncertainty through each handoff; 377 SQuAD items on which the receiver reading the clean passage reaches token-F1 $\geq .5$. AUROC is of the receiver’s first-token entropy against the sender’s semantic entropy (thresholded at its median per variant); the drop ratio is the receiver’s $P(\text{gold})$ loss from clean to removed relative to the text path’s. The two “scored directly” rows are the message-level signal: the label itself ranked against the same target, with no receiver (Section 5). Two projectors trained with uncertainty objectives are in Appendix B.

receiver sees	$P(\text{gold})$	entropy	AUROC vs sender SE	drop ratio
	clean / contra. / removed	clean / removed	clean / contra. / removed	
text	.437 / .290 / .162	1.14 / 3.03	.597 / .615 / .671	1 (def.)
text + answer + confidence word (verbal)	.253 / .193 / .171	2.88 / 2.80	.552 / .542 / .592	.30
text + answer + numeric confidence (sample agreement, e.g. “0.8”)	.258 / .197 / .170	2.93 / 2.93	.548 / .532 / .563	.32
text + all candidate answers with their sampled probabilities	.264 / .210 / .144	2.68 / 2.63	.495 / .489 / .626	.44
mapped cache	.279 / .249 / .130	1.78 / 3.16	.585 / .605 / .624	.54
deranged cache	.105 / .106 / .105	3.88 / 3.88	.536 / .488 / .564	0
zero cache	.034 / .033 / .037	6.41 / 6.18	.496 / .481 / .502	0
moment-matched random cache	.005 / .004 / .005	7.24 / 7.30	.457 / .472 / .580	0
question only	.127	3.69	.532 / .479 / .603	0
<i>message-level signal (no receiver)</i>				
confidence word, scored directly as a label	—	—	.955 / .942 / 1.000	—
numeric confidence, scored directly	—	—	.870 / .848 / .735	—

Measuring doubt on both sides. Each SQuAD item comes in three versions: the clean passage; the passage with one inserted sentence asserting a different answer; and the passage with the answer sentence removed. We keep the 377 items the receiver answers correctly from the clean text. The sender’s doubt is its semantic entropy (Farquhar et al., 2024) over ten sampled answers, and it rises as the evidence weakens (0.98, 1.18 and 1.96 on the three versions). The receiver’s doubt is its entropy at the first answer token. A handoff carries doubt to the extent that the receiver’s entropy predicts whether the sender’s is above its median (AUROC). We compare the memory with the receiver reading the passage itself and with three text messages a designer might write instead: the sender’s answer plus “confidence: high / medium / low”; plus a number; and plus every candidate answer with its sampled probability. The confidence word is computed from the sender’s *actual* semantic entropy, so the text channel gets an oracle label, the best case for it. Protocol details are in Appendix A.

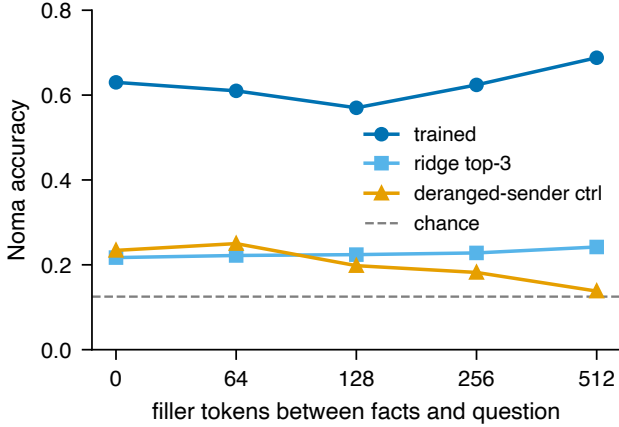
Words barely move the reader. Scored as a label, before any model reads it, the confidence word predicts the sender’s doubt almost perfectly: AUROC .955, .942 and 1.000 on the three versions (Table 5, bottom). That is mostly construction, and it is the point: the signal is in the message. The receiver that reads the message reaches .552, .542 and .592; with no message at all it reaches .532, .479 and .603. A number in place of the word does no better (.548, .532, .563), and neither does the full list of candidates with their probabilities (.495, .489, .626). This base model, under a plain completion prompt, does not turn a written label into its own doubt. The label carries the doubt. The reader drops it.

The memory does move it. Through the transplanted memory the receiver reaches .585, .605 and .624, close to reading the passage itself (.597, .615, .671), and above the confi-

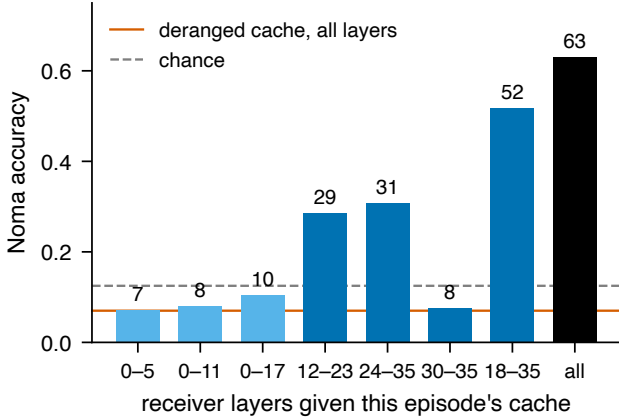
dence word on every version (Figure 4, left). The cleanest test holds the question fixed and changes only what the sender read: we remove the answer sentence, the sender’s doubt rises, and we ask, item by item, whether the receiver’s doubt rises with it. Through the memory the Spearman correlation between the two changes is .280; through the passage, .322; through the confidence word, .146; through another item’s memory, which shares the question and differs only in content, .004 (Figure 4, right). The receiver’s confidence follows what the sender saw, and it follows because of what is in the memory.

How large, and how sure. The effect is modest. Paired bootstrap intervals over passages exclude zero for the memory over the confidence word on the contradicted version ([.008, .117]) and not on the other two; reading the passage stays above the memory on the removed version ([−.091, −.002]). We had pre-registered a stronger bar for the memory, an AUROC of .70, and it was not met. What the evidence supports is that the evidence crosses and the receiver recomputes its doubt from it, about as well as from the passage and better than from any summary we could write. Whether the sender’s own belief crosses as such is a further question.

What we could not fix, and where the loss sits. The sender’s own hidden state predicts its doubt better (AUROC .739 on clean passages) than any receiver state does: through the memory .643, through the passage .650 (Appendix A). On clean passages the receiver ends up with as much of the sender’s doubt through the memory as through the text, and in the shape of the doubt, which alternatives the sender weighed, it is the same: the receiver’s mass on the sender’s candidates follows the sender’s frequencies about as well through the memory (.199) as through the passage (.193), and better than through an explicit list (.166). Two attempts to train the remaining loss away, by making the



(a) Accuracy against filler tokens between the facts and the question (8-name variant; $n = 1000$ at 0, $n = 500$ otherwise). No decay; the best value is at the training prefix length of 512.



(b) Accuracy when only the labelled band of receiver layers receives this episode’s mapped cache (the deranged episode’s cache fills the rest; Table 4). Solid line: the deranged cache in all layers; dashed: chance.

Figure 3. Distance sweep and layer-band injection.

receiver’s distribution through the memory match its distribution through the text or the sender’s own entropy, did not move it (Appendix B).

6. Portable, Within Limits

Across families. A memory is not tied to one family of models. With Qwen3-4B as the sender (a different vocabulary and different pre-training, but the same cache shape as the receiver), we align the two tokenisations by character offsets and train the same map. On the six-name binding test the receiver reads Qwen’s memory at $71.7\% \pm 4.5$ over three seeds, against 73.2% within the K2 family, and at 48.3 F1 on SQuAD. The map also runs backwards, from the 7B into the 3.7B (63.3%). It fails for K2-Horizon-0.9B (17.0%), a sibling that differs from the receiver in vocabulary, depth and head size at once; which of those differences matters

Table 6. What the message costs. Bytes per prefix token of the mapped cache (512-token SQuAD passages; “KB” = 1024 bytes). Noma: 6-name variant, $n = 300$, seed-2 projector. “filler” fills the untransmitted layers or heads with the receiver’s own cache of a neutral text of the same length. The full sweep, including zero and mean-vector fills and smaller projectors, is in Appendix G.

message	KB / token	Noma	SQuAD F1
full mapped cache, bf16	144	75.3	53.0
int4 fake-quantised	36	75.0	51.6
layers 18–35 only, filler elsewhere	72	28.7	16.1
4 of 8 heads, filler elsewhere	72	26.7	12.8

we have not isolated (Figure 5, Appendix D).

What a memory weighs. Memory is heavy (Table 6). The mapped cache is 144 KB per token, 72 MiB for a 512-token page, against about a kilobyte of text. Inside one machine that is a bargain: on one A100 the map takes 24 ms where the receiver would take 48 ms to read the page again. Across a 1 Gbps link the same memory takes about 604 ms to send. Four-bit quantisation of the mapped cache costs almost nothing (75.0% on the six-name test against 75.3, 51.6 F1 against 53.0), so a quarter of the bytes is free; cutting layers is not, even when the missing layers are filled with a structurally valid state instead of zeros (28.7%, 16.1 F1 for the second half of the network alone; Appendix G). Today a transplanted memory is worth sending inside a machine and not yet across a network. Teaching a sender to compress what it knows into kilobytes, and keep its doubt, is the problem this paper leaves open.

7. Related Work

This paper sits where three old ideas meet. Communication, in Shannon’s sense, is the reproduction at one point of a message chosen at another (Shannon, 1948); minds, in Minsky’s, are societies of smaller agents (Minsky, 1986); and a model’s uncertainty can be read from its outputs and its hidden states (Farquhar et al., 2024; Kossen et al., 2024), although models reading a stated confidence is another matter (Tian et al., 2023; Yona et al., 2024). Between language models, messages have long been text; agents have also exchanged embeddings (Pham et al., 2023), hidden states (Du et al., 2025; Peng et al., 2026), and caches, through learned fusers (Fu et al., 2025), closed-form maps within a family (Heo et al., 2026) and shared latent spaces (Dery et al., 2026; Chen et al., 2026). Audits of these channels showed that much of what looks like communication is extra computation (Cheng et al., 2026; Zhang & Emu, 2026), and the argument that a handoff in words launders the sender’s confidence (Shi et al., 2026) set the question we measure. We add the wrong-memory and wrong-sender controls, a within-item test of doubt, and the separation of what a message carries from what its reader takes.

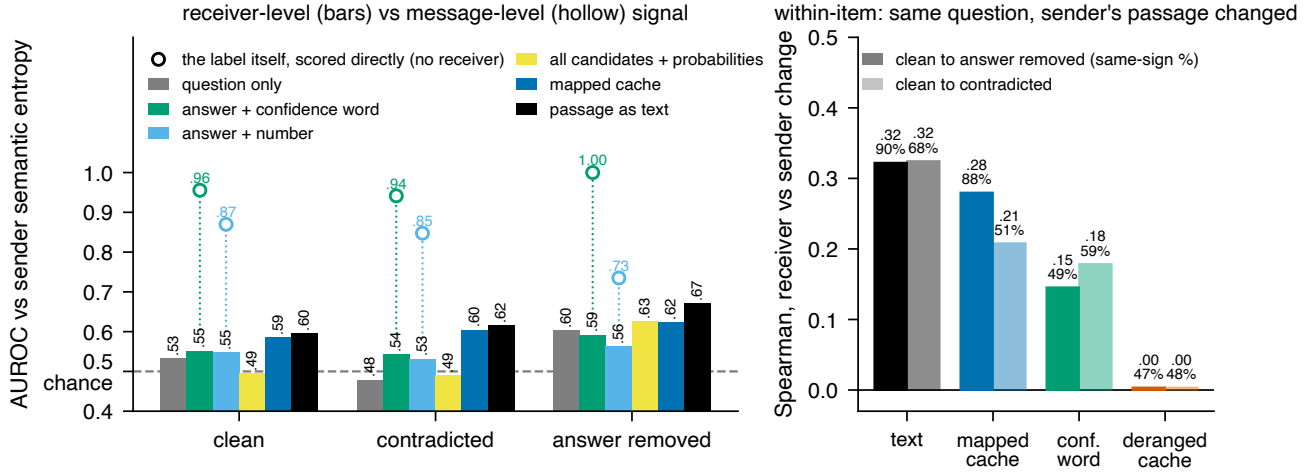


Figure 4. Left: receiver-level signal (bars: AUROC of the receiver’s first-token entropy against the sender’s semantic entropy, 377 items) for six handoffs per passage variant, with the message-level signal of the label itself drawn as a hollow marker above the word and number arms; the vertical distance from marker to bar is the read-out gap. The cache has no label and no marker. Paired, passage-grouped intervals on the three cache differences are in the text. Right: within-item intervention, the question held fixed and the sender’s passage changed; bars are the Spearman correlation between the receiver’s entropy change and the sender’s semantic-entropy change, annotated with the same-sign rate, which is bounded by the 91.5% of items on which the sender’s entropy rises (Section 5).

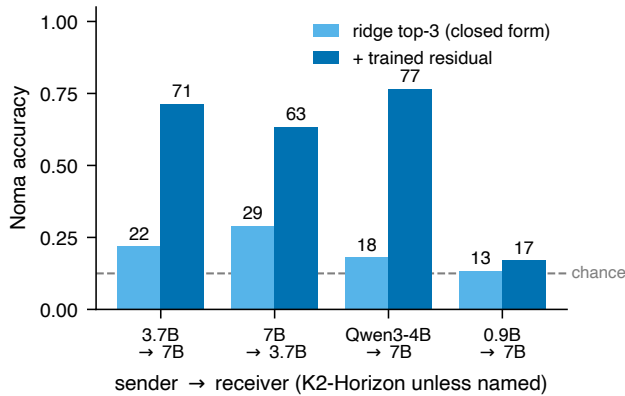


Figure 5. Noma accuracy (*project arm*) for the closed-form ridge map and the trained projector on four sender-receiver pairs ($n = 300$, 6-name variant). Dashed line: chance. The trained map crosses families (Qwen3-4B) but not a vocabulary and head-dimension change within the family (K2-0.9B).

8. Limitations

The doubt result rests on 377 items of one dataset, one pair of models and one trained map; its effect is a few hundredths of AUROC and is established on one of three passage versions. “The reader drops the label” is a property of one base-model receiver under one completion prompt; an instruction-tuned receiver, or one told to hedge, may read the label, and that is the first experiment that could overturn the result. We pre-registered an AUROC of .70 for the memory and did not reach it. The setting is the reuse of a sender’s memory of a passage, not the handoff of an agent’s reasoning or tool state in the middle of a task. The

memory is 72 MiB per page, and nothing we tried below that size keeps the layers it needs. On SQuAD the larger model is only a few points better than the smaller, so no policy for when to hand off pays there (Appendix H); the question of when a memory is worth sending needs a task with a larger gap.

9. Conclusion

One language model can read another’s memory. It reads facts it never saw, it believes what it is given, its confidence follows the sender’s evidence where a written “confidence: low” does not, and the recipe crosses model families. The memory is still too heavy to send far, and the doubt it carries is still faint. But the systems we are building are societies of models, and their members do not have to talk the way we do. The first thing to measure, for any message between them, is simple: score the message, then score the reader.

Impact Statement

This paper presents work whose goal is to advance machine learning. One consequence is specific to it: a receiver that believes whatever memory it is handed (Section 4) can be misled by a forged one. Systems that pass caches between models or devices should authenticate them as they would any other input.

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A. Uncertainty Protocol and Diagnostics

Uncertainty protocol: items and target. Following the knowledge-conflict and semantic-entropy protocols (Longpre et al., 2021; Xie et al., 2024; Farquhar et al., 2024), each SQuAD item gets three passages: *clean*, *contradicted* (one fluent sentence asserting a counter-answer, written by the 375B judge and kept only if DeBERTa-MNLI says it entails the counter-answer and not the gold: 587 of 941 survive), and *removed* (the gold sentence deleted). We keep the 377 items (344 distinct passages) on which the receiver reading the clean passage reaches token-F1 $\geq .5$. The sender’s uncertainty is its semantic entropy (SE) over 10 sampled answers (mean 0.98 / 1.18 / 1.96 over the 377 kept items on the three variants, so the sender does notice); the target is SE thresholded at its median per variant. Ten samples make SE coarse: 38 / 40 / 28 distinct values per variant, and on removed passages 22% of items sit exactly at the threshold. The receiver’s uncertainty is its entropy at the first answer token.

The four messages. Two are functions of the passage alone, $M = f(c)$, with the question unseen by the sender: *text* (the receiver reads the passage) and *mapped cache*. Two require the sender to answer the question, $M = f(c, q)$: *verbal* (the sender’s greedy answer and “confidence: high / medium / low”) and its variants with a number (the sender’s sample agreement, e.g. “0.8”, the intrinsic-confidence definition of Yona et al. (2024)) or the full candidate list (every answer cluster with its sampled frequency, e.g. “1929 (0.6); 1932 (0.3); 1935 (0.1)”). The four are therefore not one family of messages; the verbal arms have seen the question and the state arms have not, and we compare them anyway because that is the choice a system designer faces. The confidence word is an oracle label built from the sender’s *actual* SE terciles, favourable to the text channel; the sender does not verbalise its own confidence. The terciles are taken over all 587 items per variant and the target median over the 377 kept items, which is why the label’s own AUROC is below 1 on two variants (Appendix A.1). The receiver’s prompt is a plain completion—“A smaller model that read the passage answered: X (confidence: *high*).” followed by the question—with no chat template, no few-shot examples and no instruction to hedge; we did not ablate it, and Section 8 says what that leaves open.

Controls and metrics. Content-free controls: a *deranged cache* (another item’s mapped cache), a *zero cache*, a *moment-matched random cache*, and *question only*. Each handoff is scored at two levels against the same target u_S (sender SE above its median): the *message-level signal*, the AUROC of a fixed function of the message alone (a confidence word’s rank; for the cache, a linear probe on the mapped message), and the *receiver-level signal*, the AUROC of the receiver’s first-token entropy after reading it. Their difference is the read-out gap: a difference of two AUROCs, not an information decomposition, and the cache’s two levels use different instruments, so the gap is comparable within an arm, not across arms. The drop ratio is a handoff’s decrease in the receiver’s $P(\text{gold})$ from clean to removed divided by the text path’s decrease; it is a relative drop in confidence, not a fraction of uncertainty information retained, and the paths start from different confidence levels. Values are in Table 5.

Capacity, stated correctly. A message that is the argmax of the sender’s belief p_S plus a k -level label takes at most $|Y| k$ values, so it carries at most $\log |Y| + \log k$ nats about p_S ; a state message is not so bounded, although what the receiver recovers from it is bounded by the map and by its reading. This limits information about the *distribution* p_S (which alternatives, how spread), not a detector’s AUROC against a thresholded target: a three-level label reaches AUROC .955 against the median split (Appendix A.1). An earlier draft of this paper deduced a lower receiver-level AUROC for the text arm from this “log 3 nats” bound; that deduction was wrong and is withdrawn. The bound is not what limits the text arm here, and no measurement in this paper tests it. *Why recovering the belief would matter.* If the receiver’s recovered belief $\tilde{b}$ is within total variation ε of the sender’s b and one-step rewards lie in $[0, R_{\max}]$, then acting on $\tilde{b}$ instead of b loses at most $2R_{\max}\varepsilon$ in expected reward. That is what a decision test should measure; this paper measures the read-out gap and the tracking of u_S , not regret.

Controls, per variant. The content-free controls do not sit at one level. On clean and contradicted passages they lie at .46–.54 (question only .532 / .479; deranged .536 / .488; random .457 / .472; zero .496 / .481) against the cache’s .585 / .605. On removed passages question-only alone gives .603, the random cache .580 and the deranged cache .564 against the cache’s .624: the question already carries most of the signal when the answer is gone, and question-only there beats the word (.592). The cache’s increment over question-only is therefore +.053 on clean, +.126 on contradicted and +.021 on removed passages, and only the middle one is established.

Behaviour under contradiction and removal. When the answer is removed, the cache path loses 54% of the confidence the text path loses and its entropy rises by 73% of the text path’s rise. Under a contradiction the text path is swayed slightly more by the inserted sentence than by the original ($P(\text{counter}) .32 > P(\text{gold}) .29$), while the cache path keeps the original ahead (.25 > .22); across all conditions the cache path is less confident than the text path on clean passages ($P(\text{gold}) .28$ vs .44), so a receiver that treats the two channels as equivalent would under-trust the cache (Figure 9). What this section cannot say is why the ordering is what it is; that is the next section’s job.

A.1. Where the loss occurs

(a) The label carries the signal; this receiver does not read it. The confidence word is built from the sender’s own SE terciles, so scored as a label it predicts the binary target almost perfectly without any receiver: AUROC .955 / .942 / 1.000 on the three variants, the numeric label .870 / .848 / .735 (Table 5, bottom). These numbers are a property of the construction—a tercile split over all 587 items against a median split over the 377 kept items—and the 1.000 on removed passages is a degeneracy: 22% of items tie at the threshold and the tercile boundary coincides with it. The audit is a sanity check, not a finding; the finding is the other bar: the receiver’s first-token entropy after reading that word reaches .552 / .542 / .592. The text channel is not capacity-limited here; this base-model receiver, under one completion prompt, does not turn a label into its own spread. That is a statement about one reader and one prompt format; a receiver instructed to hedge, given few-shot examples, or instruction-tuned could erase the contrast, and we did not test one. It is also why the capacity argument of our earlier draft (the box above) cannot be the mechanism of the verbal arm’s shortfall.

(b) Within-item intervention. Cross-item correlations admit a common cause: a hard question makes both models uncertain whether or not anything crosses. The variants allow a paired test with the question held fixed and only the sender’s passage changed (clean → removed; the sender’s SE rises by 0.98 on average). Per item, the Spearman correlation between the receiver’s entropy change and the sender’s SE change is .280 through the mapped cache, .322 when the receiver reads the text, .146 through the confidence word, and .004 through a cache from another item, which shares the question and rules out the common cause (Figure 4, right). The same-sign rates (88 / 90 / 49 / 47%) are weaker evidence than they look: the sender’s SE rises on 91.5% of items, so any arm whose entropy simply rises on removal scores near that ceiling and an arm with no mean shift sits at a coin flip; same-sign measures the mean shift, the Spearman the per-item covariation. Under a contradiction the effect is smaller (cache .208, text .325, word .179, deranged .004). The receiver’s confidence moves with what the sender saw, and it moves because of the cache’s content. It licenses no more than that: the text path tracks at least as well, so the parsimonious reading is that the evidence crosses and the receiver recomputes its own uncertainty from it.

(c) Shape of the belief. Sender SE and receiver first-token entropy are different measurements; a diagnostic closer to the belief itself is the *shape*: per item with at least two sampled answer clusters, the Spearman correlation between the sender’s cluster frequencies and the receiver’s first-token mass on those candidates. On clean passages ($n = 323$) it is .199 through the mapped cache, .193 from the text, .166 from the explicit candidate list, and .052 / .042 for the deranged cache and question only: there the receiver recovers the sender’s alternatives about as well from the cache as from the passage, and no better from a list of them. Under a contradiction ($n = 351$) the order reverses (cache .153, list .201, text .234), and on removed passages every arm is near zero. No interval was computed for these correlations, and they are small enough that the diagnostic separates arms from the controls, not from each other.

(d) Four-point probes. The same standardised logistic probe ($C = 0.05$, five passage-grouped folds, no inner selection; the semantic-entropy-probe protocol of Kossen et al., 2024) for the binary sender-SE target at four points of the chain, on inputs of 2560 / 2048 / 6144 / 4096 dimensions and on the 587 items (the full set before the F1 filter, so these AUROCs are not directly comparable with Table 5). Clean / removed: the sender’s own last-prompt-token state, averaged over layers 12 / 18 / 24, gives .739 / .706; the mapped message at its last prefix position, averaged over layers and heads, .557 / .579 (with the question-only receiver state concatenated, .563 / .603); the receiver’s state after reading through the mapped cache .643 / .563; after reading the text .650 / .654; the question alone .530 / .591. Three things follow. The sender holds more about its own uncertainty than any receiver state recovers. On clean passages the receiver reaches .643 through the cache, above the .557 that the probed message carries, so the message carries more than a one-position probe reads, and the cache leaves the receiver with as much as the text does (.643 vs .650); on removed passages the receiver (.563) falls below the probed message with the question (.603) and below the text (.654). These probes therefore do not localise the loss. Two caveats bound them further. The message-level probe sees one position of 512, so .557 / .579 bounds what one position carries, not what the map preserves; a pooled read-out of the whole prefix is the obvious next probe. And the sender probe saw the question as well as the passage, while the mapped message is passage-only, so sender-minus-message mixes question conditioning with map loss.

B. Training Objectives That Did Not Recover the Sender’s Spread

Adding $KL(\text{text-path} \parallel \text{cache-path})$ on the receiver’s next-token distributions to the projector’s objective (weight 1, 1,000 steps) raises the cache path’s entropy everywhere (clean 2.38 vs 1.78) without making it track the sender better (AUROC .590 / .589 / .611; drop ratio .56 vs .54) and costs 2 F1. Matching the receiver’s own text-path distribution is the wrong target: what is lost is the *sender’s* spread, which the receiver’s text path does not contain either. A sender-side target—matching the

Table 7. Two projectors trained with an uncertainty objective (weight 1, 1,000 steps), scored exactly as in Table 5.

receiver sees	$P(\text{gold})$	entropy	AUROC vs sender SE	drop ratio
	clean / contra. / removed	clean / removed	clean / contra. / removed	
mapped cache (reference, Table 5)	.279 / .249 / .130	1.78 / 3.16	.585 / .605 / .624	.54
mapped cache, KL-trained projector	.272 / .241 / .118	2.38 / 3.76	.590 / .589 / .611	.56
mapped cache, sender-entropy-matched projector	.247 / .222 / .107	2.23 / 3.64	.596 / .587 / .627	.51

receiver’s per-token entropy through the cache to the sender’s own per-token entropy on the training text—was the second attempt (weight 1, 1,000 steps): binding accuracy holds (72.3%; Appendix C), SQuAD F1 falls to 45.3, and the uncertainty measures do not move (AUROC .596 / .587 / .627; drop ratio .51). Entropy matching is not belief matching; the untested fix is a separate reader $r(m, q)$ trained on the sender’s candidate distribution.

C. Ablations on the 6-Name Variant

All runs: sender K2-Horizon-3.7B, receiver K2-Horizon-7B, colour attribute, 6 names, $n = 300$ episodes, chance 12.5%. Accuracy in percent. The two 1,000-step rows with an added uncertainty term (KL, sender-entropy matching) use the full 151 M-parameter projector.

Table 8. Projector ablations on the 6-name Noma variant.

projector	project	derange	follow
ridge top-3 (closed form)	22.0	10.7	30.3
+ residual, 300 steps	56.3	6.3	53.7
+ residual, 1500 steps, seed 0	71.3	4.3	71.7
+ residual, 1500 steps, seed 1	73.0	3.0	74.0
+ residual, 1500 steps, seed 2	75.3	2.3	76.0
+ residual, 1500 steps, ridge frozen	59.0	7.0	59.3
+ residual, 1500 steps, aligned-layer ridge	32.7	8.0	34.7
+ residual, 1000 steps, bottleneck 32	64.3	5.7	62.3
+ residual, 1000 steps, bottleneck 16	61.3	7.7	60.0
+ residual, 1000 steps, + KL(text cache)	66.0	5.3	63.3
+ residual, 1000 steps, + sender-entropy matching	72.3	3.0	73.3

D. Other Attributes, Reversed Roles, and Other Pairs

Table 9. Binding transfer with city and animal attributes ($n = 500$, 8 names) and with the roles reversed (7B sender, 3.7B receiver; 6 names, $n = 300$).

setting	projector	none	project	derange	follow
city	ridge top-3	14.2	19.8	11.8	18.4
city	trained	14.2	61.8	6.4	60.2
city	deranged-sender ctrl	14.2	15.4	11.4	18.6
animal	ridge top-3	12.2	12.8	12.0	11.2
animal	trained	12.2	67.2	5.2	64.6
animal	deranged-sender ctrl	12.2	12.4	12.8	9.8
reversed	ridge top-3	11.7	29.0	12.3	31.3
reversed	trained	11.7	63.3	5.7	63.0

The channel is not one-way: with the roles swapped (7B sender, 3.7B receiver) the ridge map gives 29.0% and the trained projector 63.3%, with the deranged cache at 5.7% and a 63% follow rate—the same pattern at slightly lower accuracy. With Qwen3-4B as the sender (36 layers, 8 KV heads, head dimension 128—matched KV geometry—but a different vocabulary and pre-training), receiver positions are aligned to sender tokens by character offsets and the same recipe is run: top-3 ridge

Table 10. Four sender–receiver pairs, same recipe. Noma 6-name variant, $n = 300$, chance 12.5%; percentages. “proj.” is the *project* arm, “der.” the *derange* arm, “follow” the follow rate. SQuAD F1 is for the trained projector ($n = 300$; text ceiling 54.7). K2-Horizon unless named. Qwen3-4B row: proj. and F1 are mean $\pm$ sd over three training seeds (76.7 / 70.3 / 68.0; F1 48.0 / 51.9 / 44.9); der. is the worst seed, follow is seed 0. The raw arm is undefined for K2-0.9B (head dimension 64).

sender → receiver	raw	ridge top-3			+ trained residual			F1
		proj.	der.	follow	proj.	der.	follow	
3.7B → 7B	14.0	22.0	10.7	30.3	71.3	4.3	71.7	53.0
7B → 3.7B	11.7	29.0	12.3	31.3	63.3	5.7	63.0	—
Qwen3-4B → 7B	15.7	18.0	13.0	17.0	71.7 $\pm$ 4.5	$\leq$ 4.7	80.3	48.3 $\pm$ 3.5
0.9B → 7B	n/a	13.3	13.0	10.7	17.0	14.0	11.3	24.2

18.0% (raw injection 15.7%, deranged 13.0%), trained projector 76.7 / 70.3 / 68.0% over three training seeds (mean 71.7, sd 4.5; deranged $\leq$ 4.7%, follow rate 80% for seed 0), SQuAD F1 48.0 / 51.9 / 44.9 (mean 48.3) against 54.7 for the receiver reading the text. The linear map alone does nothing across families. The recipe fails for K2-Horizon-0.9B, a same-family sender with a different vocabulary and head dimension 64: the trained projector reaches 17.0% (deranged 14.0%, follow 11.3%; content transfer 3 points) and 24.2 F1 on SQuAD. K2-0.9B differs from the receiver in vocabulary, layer count and head dimension at once; which difference matters is untested, and cross-head maps exist in other work, so the failure is of this parameterisation, not of the pair.

E. Different Contexts: HotpotQA

Table 11. HotpotQA bridge questions, $n = 400$. Sender and receiver each hold one of the two supporting paragraphs; the receiver always reads its own paragraph and the question. Token-F1; the mapped cache is mean $\pm$ sd over three training seeds (37.9 / 39.7 / 42.1), the deranged cache their range.

receiver additionally gets	F1
nothing (own paragraph + question)	30.1
the sender’s paragraph as text	45.8
the sender’s paragraph as a mapped cache	39.9 $\pm$ 2.1
a deranged mapped cache	29.9–32.1
the sender’s paragraph through the ridge map	26.3
a zero cache	20.9

This is the LCF-X setting of Rossi et al. (2026). The receiver with its own paragraph and the question reaches F1 30.1; with both paragraphs as text 45.8; with its own paragraph plus the sender’s mapped cache 37.9 / 39.7 / 42.1 over the three training seeds (mean 39.9, sd 2.1)—63% of the gap—while the deranged cache gives 29.9–32.1 (no help), the ridge map 26.3 and a zero cache 20.9. The channel carries the *other* paragraph, and only the trained map does.

F. Distance Sweep and Layer Bands

Table 12. Noma accuracy (*project* arm) versus filler tokens between facts and question. 8-name variant; $n = 1000$ at pad 0, $n = 500$ otherwise.

projector	0	64	128	256	512
ridge top-3	21.7	22.2	22.4	22.8	24.2
trained	63.0	61.0	57.0	62.4	68.8
deranged-sender ctrl	23.4	25.0	19.8	18.2	13.8

Table 13. Every compression of the seed-2 projector’s mapped cache. KB per token is for 512-token SQuAD passages; low-rank truncation and mean-vector fill have a per-sequence overhead, so their per-token size depends on sequence length (for the 29-token Noma facts, rank 32 exceeds the full cache). Noma: 6-name variant, $n = 300$, percent; SQuAD: $n = 300$, token-F1. “filler” rows fill the untransmitted layers or heads with the receiver’s own cache of a neutral text of the same length.

spec	KB / token	Noma project	Noma derange	SQuAD F1
none (full, bf16)	144	75.3	2.3	53.0
int8	72	76.0	2.7	52.1
int4	36	75.0	3.7	51.6
rank 32	78.7	75.0	2.3	45.3
rank 16	39.3	75.0	2.3	29.6
rank 8	19.7	51.7	7.0	17.2
layers 18–35, zeros	72	24.3	9.7	8.3
layers 18–35, zeros, int8	36	24.7	9.7	7.8
4 heads, zeros	72	25.7	8.7	8.2
2 heads, zeros	36	22.3	9.7	5.9
layers 12–35, mean fill	96.4	35.0	11.0	28.8
layers 18–35, mean fill	72.7	33.7	11.7	21.4
layers 18–35, mean fill, int4	18.7	33.3	11.7	21.1
layers 24–35, mean fill	48.9	29.3	12.0	14.5
4 heads, mean fill	72.7	23.0	11.3	15.3
layers 12–35, filler	96	30.7	11.7	18.9
layers 18–35, filler	72	28.7	12.0	16.1
layers 18–35, filler, int4	18	30.0	11.7	16.9
layers 24–35, filler	48	25.3	10.0	13.3
4 heads, filler	72	26.7	10.0	12.8
projector bottleneck 32 (70 M params)	144	64.3	5.7	53.7
projector bottleneck 16 (62 M params)	144	61.3	7.7	51.6

G. Compression Sweep

A projector with bottleneck 32 has 70 M parameters and keeps SQuAD F1 (53.7) while losing 9 points on the binding test (64.3%); bottleneck 16 (62 M) gives 61.3% and 51.6 F1. The filler rows are the structural null of Section 6: the receiver’s own cache of a neutral text is a valid, episode-free state, so a collapse under it is missing content rather than broken attention.

H. When Should the Small Model Hand Off?

Table 14. Handoff policy on 600 SQuAD items. Top: AUROC of each trigger at predicting the sender’s own failures ($F1 < 0.5$). Bottom: F1 of each arm when every item is handed off (rate 1).

sender (3.7B) alone: F1 52.2, wrong on 35.5% of items	
trigger	AUROC
probe on the sender’s hidden state	.749
semantic entropy (10 samples)	.689
sample disagreement	.683
max-probability	.582
entropy	.584
receiver (7B) sees	
F1 at rate 1	
the passage as text	55.7
the mapped cache	51.8
the sender’s answer + confidence word	47.0

On 600 SQuAD items the 3.7B alone reaches F1 52.2 and gets 35.5% of items wrong ($F1 < 0.5$). A logistic probe on its pre-action hidden state (last prompt token, layers 12/18/24; the probe of Kossen et al., 2024 with a failure label; five-fold, not grouped by passage, $C = 0.1$) predicts those failures with AUROC .749, better than its own semantic entropy (.689),

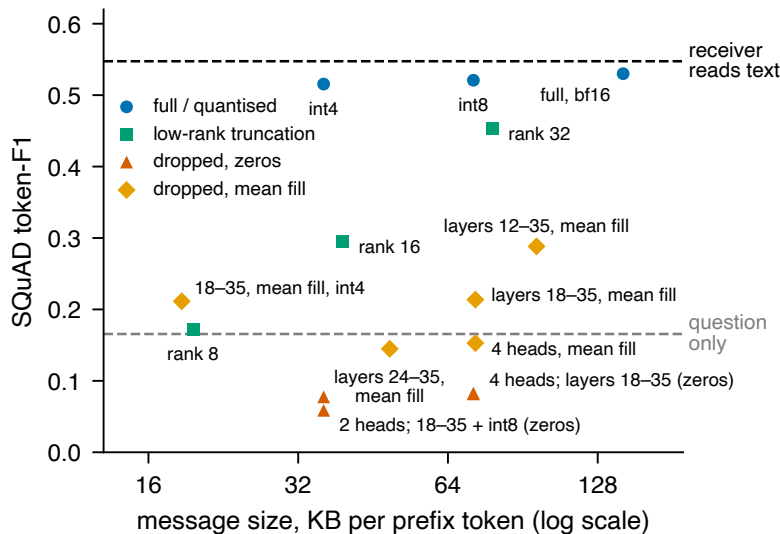


Figure 6. Message size against SQuAD F1 for every compression of the mapped cache in the zero- and mean-fill sweep ($n = 300$). Quantisation is free; low-rank truncation and dropping layers or heads are not. The two “zeros” points at 72 and 36 KB each hide two coincident specs.

sample disagreement (.683), max-probability (.582) or entropy (.584) (Table 14). But the 7B reading the passage reaches only 55.7, the mapped cache 51.8 and the verbal handoff 47.0, so no trigger makes a handoff pay at equal latency (Figure 7): the probe-triggered text handoff climbs from 52.2 to 55.7 as the handoff rate goes to 1, the cache handoff stays flat at 52, and only an oracle trigger gains (56.4 at rate .4). On this task the receiver is not enough better than the sender for a handoff of any kind to be worth it; the probe result stands, the policy result waits for a task with a real gap.

I. Additional Figures

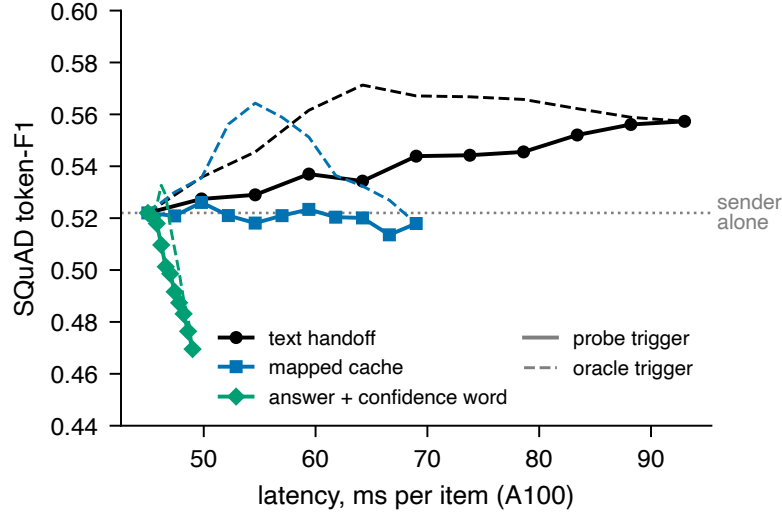


Figure 7. F1 against latency per item as the handoff rate goes from 0 (sender alone, 45 ms) to 1, for the three handoff arms under the probe trigger (solid) and an oracle trigger that hands off exactly the sender’s failures (dashed). 600 SQuAD items. Only the oracle gains; the probe-triggered text handoff climbs to the receiver’s own 55.7, the cache handoff stays flat.

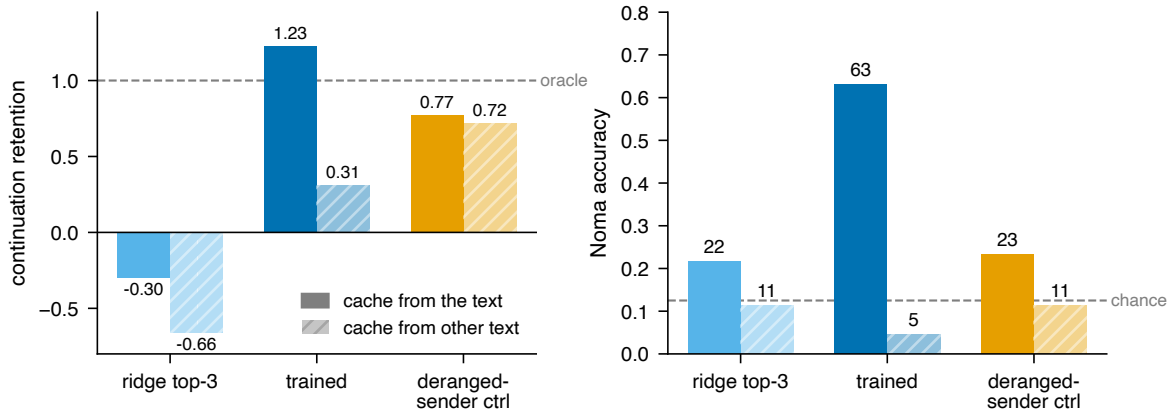


Figure 8. Retention and the binding test disagree about the deranged-sender control. Left: continuation retention on 128 fresh held-out sequences for the three projectors, with the sender reading the true prefix (solid) or a swapped one (hatched). Right: Noma accuracy for the same three projectors ($n = 1000$, 8-name variant), *project* arm (solid) and *derange* arm (hatched). Retention ranks the deranged-sender projector at 0.77, most of which survives a swapped sender; the binding test puts it at chance.

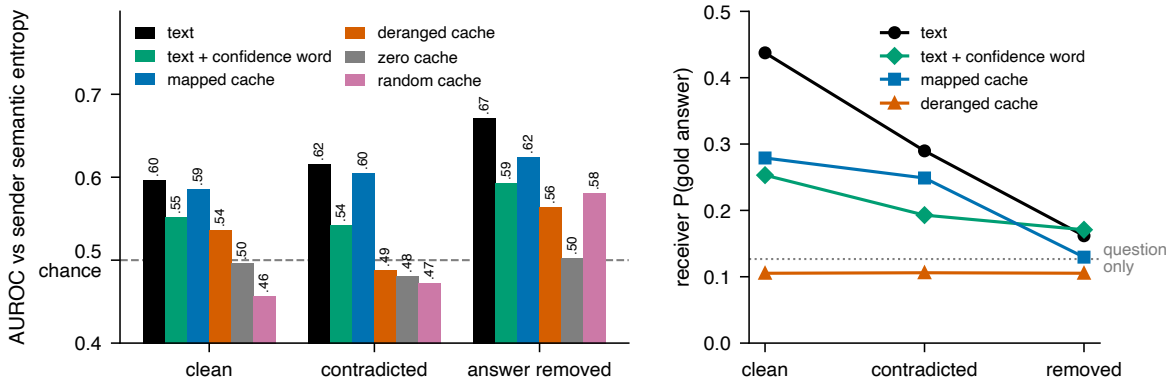


Figure 9. Left: AUROC of the receiver’s first-token entropy against the sender’s semantic entropy, per handoff and passage variant (377 items), including the zero and moment-matched random caches. Right: the receiver’s $P(\text{gold})$ from clean to contradicted to removed passages. The cache path is less confident than the text path on clean passages and keeps the original answer ahead under a contradiction.